DRUMPANEL · USER REQUIREMENTS

DrumPanel user requirements

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DOCUMENT CONTROL

Document ID	DMP-URS-001	Version	0.1
Status	Draft	Date	2026-10-24
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/drumpanel		
Source	docs/03a-user-requirements.md @ de21d74		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First user requirements (kit 1.8 upgrade), drawn from the users, stakeholders, constraints and co-design notes of DMP-PRB-001; desk draft to review with a co-design partner

DrumPanel user requirements

What the people who open drums, build the tools and buy the sheet need, written from their side. Fifteen needs, eight of them High. They are drawn from the users and stakeholders tables of DMP-PRB-001 v0.4 and its constraints; every need traces to product requirements in DMP-PRS-001 (matrix: DMP-RTM-001), and the same needs scored the concept options (DMP-OPT-001). This is a desk draft: a cooperative of drum artists and a vocational welding school review it before TRL 4 (co-design candidates in DMP-PRB-001; nothing agreed). Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Safety	No fire, no smoke	Drums are opened without burning anything, so there is no smoke from paint and residue in the yard or the neighbourhood	High	Metal artists and apprentices; neighbours (PRB Table 1, Table 2)
UR-002	Safety	Know the drum is safe	Before anyone cuts, we know the drum holds no flammable vapour, from a simple check we trust	High	Apprentices; cooperative; vocational school (PRB Table 1)
UR-003	Performance	Heads off, cold and quick	Both heads come off a drum cold, in a few minutes, as flat discs we can use	Medium	Metal artists and apprentices
UR-004	Performance	Clean, straight cuts	The side opens along a straight line and the cut edges are clean, so the sheet is ready for art	Medium	Metal artists
UR-005	Performance	Flat sheet	The sheet comes off flat enough to mark out and cut designs without hours of hammering	High	Metal artists
UR-006	Usability	Hand power that does not wear us out	The tools work by hand, with no electricity, and an apprentice can work them all day	High	Apprentices; workshops without reliable power (PRB operating environment)
UR-007	Performance	Enough sheet for a workshop cluster	Two people prepare enough drums in a working day to keep a cluster of workshops in sheet	Medium	Artisan cooperative
UR-008	Maintenance	Made and mended locally	A town fabricator with a welder and a lathe can build and repair the set from drawings and commodity parts	High	Local fabricators (PRB Table 1)
UR-009	Cost	Affordable for a cooperative	A cooperative or a fair trade partner can fund one shared set	Medium	Artisan cooperative; fair trade buyers
UR-010	Safety	No trapped fingers	Nobody's fingers can reach a roll, disc, gear or chain while the tools are worked	High	Apprentices; cooperative
UR-011	Safety	No heavy lifts	Nobody lifts more than they safely can, and nobody tips a drum full of water	Medium	Apprentices
UR-012	Performance	Useful sheet from every drum	Every drum gives sheets of a known, useful size and the heads as discs	High	Metal artists; fair trade buyers
UR-013	Environment and durability	Oil kept out of the ground	Residue and purge water are caught, settled and reused, and the oil goes to a recycler, never onto the ground	Medium	Communities near the yards; drum collectors and recyclers (PRB Table 2)
UR-014	Usability	A method we can follow from pictures	Every step, from accepting a drum to stacking the sheet, can be followed	Medium	Cooperative; vocational school

ID	Category	Title	Need	Priority	Source
			from pictogram sheets in our own language		
UR-015	Safety	Only safe drums come in	Drums that held fuel, solvent, chemicals or unknown contents never enter the process	High	Drum collectors and recyclers; cooperative (PRB constraints)

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.